
WEAK FORM RECOVERY OF HESTON TYPE STOCHASTIC DYNAMICS

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ABSTRACT

Estimating the coupled drift, diffusion, and leverage structure of a stochastic-volatility model directly from a price path is an unresolved inverse problem: Kramers–Moyal increment estimators amplify sampling noise as the step shrinks, weak-form SINDy has not been extended to coupled two-dimensional diffusions or to the return–variance cross-variation producing leverage, and Heston calibration typically relies on option-implied surfaces rather than the physical-measure path. We extend the spatial weak-form Galerkin framework to the Heston model: variance increments, squared variance increments, squared price increments, and their cross-product are projected onto shared Gaussian kernels in variance space, giving one LASSO regression that jointly recovers mean reversion κ , long-run variance θ , vol-of-vol ξ , and leverage correlation ρ , with a drift-informed bias correction analogous to scalar-SDE diffusion debiasing. Across 30 daily-observed Heston simulations, ξ , ρ , and $\rho\xi$ are recovered with median errors under 2%. Applied to S&P 500 data spanning the 2007–2010 crisis, the method recovers negative leverage consistent with the documented equity leverage effect, and a 50-stock Indian panel shows the same sign under several independent variance proxies.

Keywords weak-form regression · stochastic SINDy · Heston model · leverage effect · LASSO · variance proxies · errors in variables

1 Introduction

A central challenge in quantitative finance is the inverse problem: given an observed price path, can we recover the stochastic dynamics that generated it? For a stochastic-volatility model this question has two parts. The generator must be estimated from noisy increments, and the variance state must first be constructed because it is not observed directly. A recovery can therefore change when the variance proxy or its time indexing changes, even when the regression method remains fixed.

Sparse Identification of Nonlinear Dynamics (SINDy) estimates governing equations from a candidate function library [10], with model-selection and implicit-form extensions [23, 20] and a growing convergence theory that characterizes

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when the underlying sparse regression recovers the true support [31]. Robustness to experimental noise has also been pursued through physically constrained symbolic regression [27]. Stochastic and weak-form variants replace pointwise differentiation with moment or integral relations, which can be more stable for noisy paths [8, 28, 24]. The useful idea is simple: instead of estimating a derivative at every observation, we average increments against smooth test functions. The averaging improves numerical stability while retaining information about the drift and diffusion. A related operator-theoretic line of work estimates the infinitesimal generator itself through Koopman/EDMD approximations [21]; unlike that approach, the weak-form regression used here returns explicit symbolic coefficients rather than a numerical operator approximation, following the general stochastic-process framework of [26]. We apply this construction to the Heston model [19] and add a weak cross-variation target for the leverage effect.

The paper addresses four connected questions:

1. Can one shared weak design recover variance drift, diagonal diffusion, and return–variance cross-diffusion?
2. How does additive state noise affect the leverage coefficient $\rho\xi$, once uncertainty is measured over repeated noise draws?
3. Can a one-sided variance filter reduce proxy noise without making generator recovery worse?
4. Which conclusions remain stable when the method is applied to the S&P 500 and to several variance proxies for Indian equities?

These questions define the scope of the paper. The primary Heston library $\{1, v\}$ is specified in advance, so the main result concerns coefficient recovery inside that structure. The empirical estimates use same-day variance proxies and are interpreted as contemporaneous associations. We do not claim a real-time state estimator, causal identification, option-pricing calibration, or trading performance.

Reproducibility. All recovery fits and manuscript figures are implemented in the single notebook-style script `weak_sindy_project.py`; every numerical table is written to CSV. The supplied Indian daily proxy panel is produced upstream from the archived one-minute Parquet data. The accompanying reproducibility package records the fixed seeds, software versions, input hashes, pipeline settings, repeat-level results, and plotted summaries.

2 Methodology

This section connects the Heston parameters to quantities that can be estimated from discrete data. We first state the model, then identify the relevant increment moments, and finally show how the four weak targets are fitted with one shared numerical pipeline.

2.1 Heston dynamics

Let $x_t = \log S_t$. Under the physical-measure interpretation used here, the log price and its instantaneous variance satisfy

$$dx_t = \left(\mu - \frac{1}{2}v_t\right)dt + \sqrt{v_t}dW_{x,t}, \quad (1)$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_{v,t}, \quad (2)$$

$$\mathbb{E}[dW_{x,t}dW_{v,t}] = \rho dt. \quad (3)$$

Here κ is the speed of mean reversion, θ is the long-run variance level, ξ is the volatility of variance, and ρ is the instantaneous correlation between price and variance shocks. These parameters enter the instantaneous covariance matrix

$$a(v) = \begin{pmatrix} v & \rho\xi v \\ \rho\xi v & \xi^2 v \end{pmatrix}. \quad (4)$$

The usual Feller condition $2\kappa\theta \geq \xi^2$ is sufficient for strict positivity of the CIR variance process, and is restated as Assumption H1 below, where it also supplies the geometric ergodicity that the consistency and normality results of Section 2.4 require. The off-diagonal entry is especially important here: its slope is $\rho\xi$. A negative slope means that negative price shocks tend to coincide with positive variance shocks, which is the equity leverage effect in this diffusion model [7, 11]. The affine structure of $b(v)$ and $a(v)$ places the Heston model inside the broader affine jump-diffusion class studied by [13], whose transform methods and the exact simulation scheme of [9] we use only for generating synthetic ground-truth trajectories; the recovery pipeline itself uses the simpler Euler–Maruyama discretization of [22], since consistency of the weak estimator (Theorem 2.2) does not require exact simulation.

2.2 Generator and conditional-moment interpretation

The generator is the object we want to recover because it contains both the deterministic drift and the local covariance. For a diffusion with drift b and covariance matrix a , it is

$$\mathcal{L}f(z) = b(z)^\top \nabla f(z) + \frac{1}{2} \text{tr}(a(z) \nabla^2 f(z)). \quad (5)$$

The reason this helps is that the first two local conditional moments identify the same two components:

$$b_i(z) = \lim_{\tau \downarrow 0} \mathbb{E} \left[\frac{Z_i(t+\tau) - Z_i(t)}{\tau} \middle| Z_t = z \right], \quad (6)$$

$$a_{ij}(z) = \lim_{\tau \downarrow 0} \mathbb{E} \left[\frac{\Delta Z_i \Delta Z_j}{\tau} \middle| Z_t = z \right]. \quad (7)$$

Directly estimating these limits from one path is unstable. The spatial weak form instead aggregates many increments against Gaussian state-space kernels. The construction follows [14]; our extension adds price increments and the return–variance cross-variation needed for leverage, and establishes that the coupled two-dimensional projection remains unbiased, consistent, and asymptotically normal despite the correlated driving noise ρ shared by the two coordinates.

2.3 Standing Assumptions

The theoretical results below rest on two assumptions, adapted from the scalar-diffusion framework of [14] to the coupled Heston pair (x_t, v_t) .

Assumption H1 (Feller ergodicity of the variance marginal). The Feller condition $2\kappa\theta \geq \xi^2$ holds, so that v_t almost surely stays strictly positive and the scalar CIR process $dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_{v,t}$ possesses a unique stationary law $\pi(v) = \text{Gamma}(2\kappa\theta/\xi^2, \xi^2/2\kappa)$ and is geometrically ergodic: there exist $C > 0$, $\rho_0 \in (0, 1)$ such that $|\mathbb{E}^v[g(v_t)] - \int g d\pi| \leq C\|g\|_\infty \rho_0^t$ for every bounded measurable g [15, 12].

Assumption H2 (Regularity and library completeness). The drift and diffusion coefficients of v_t lie in the span of the library $\Theta(v) = [1, v]$, i.e. $b_v(v) = \kappa\theta - \kappa v$ and $a^{vv}(v) = \xi^2 v$ exactly. The kernels K_j are bounded, Lipschitz, and integrable against π . The log-price drift $b_x(v) = \mu - v/2$ and covariance entries $a^{xx}(v) = v$, $a^{xv}(v) = \rho\xi v$ are affine in v , so that all four weak targets defined below are exactly representable by the same design matrix A .

A structural feature of the Heston system makes Assumption H1 easier to invoke than in a generic bivariate diffusion: v_t is *autonomous*. Its own dynamics do not depend on x_t , so the triangular (cascade) structure of the system means that ergodicity of the full pair (x_t, v_t) reduces to ergodicity of the scalar CIR marginal, which is classical [15, 12, 18]. This is what licenses localizing the kernels K_j in v alone rather than in the joint (x, v) state space.

2.4 Weak Projection of the Coupled Flow

Let $\{(x_{t_n}, v_{t_n})\}_{n=0}^N$ be observations at $t_n = n\Delta t$. Writing $b_v(v) = \kappa(\theta - v)$, $b_x(v) = \mu - v/2$, and letting (ξ_n^x, ξ_n^v) be i.i.d. across n , standard bivariate normal pairs with $\text{Corr}(\xi_n^x, \xi_n^v) = \rho$ and $(\xi_n^x, \xi_n^v) \perp \mathcal{F}_{t_n}$, the Euler–Maruyama discretization of the coupled system is

$$\Delta x_n = b_x(v_{t_n})\Delta t + \sqrt{v_{t_n}} \xi_n^x \sqrt{\Delta t}, \quad (8)$$

$$\Delta v_n = b_v(v_{t_n})\Delta t + \xi \sqrt{v_{t_n}} \xi_n^v \sqrt{\Delta t}. \quad (9)$$

Multiplying by the spatial kernel $K_j(v_{t_n})$ and summing over the trajectory decomposes each of the four weak targets into a deterministic signal term (an ergodic average of a library function against K_j) and a stochastic remainder driven by (ξ_n^x, ξ_n^v) . Because $K_j(v_{t_n})$ depends on the state at t_n alone, it is $\mathcal{F}_{t_n}$ -measurable, and because the Euler–Maruyama shocks are independent of $\mathcal{F}_{t_n}$ this construction inherits the state-weighting argument of [14]: correlating the two coordinates of the driving noise through ρ does not reintroduce the temporal endogeneity bias that a time-indexed test function would produce, because unbiasedness is proved term-by-term via the tower property (Theorem 2.1 below), and that argument never uses independence between ξ_n^x and ξ_n^v themselves — only their independence from the filtration up to t_n .

Theorem 2.1 (Unbiasedness of the coupled spatial projection). *Under Assumption H2, the stochastic component of each weak target $B_j^{(b)}$, $Q_j^{(vv)}$, $Q_j^{(xx)}$, $Q_j^{(xv)}$ has zero conditional expectation given $\mathcal{F}_{t_n}$ at every step n , and hence unconditional expectation zero. Consequently*

$$\mathbb{E}[B^{(b)}] = Ac^*, \quad \mathbb{E}[Q^{(vv)}] = Ad_{vv}^* + \mathcal{O}(N\Delta t^2), \quad \mathbb{E}[Q^{(xx)}] = Ad_{xx}^* + \mathcal{O}(N\Delta t^2), \quad \mathbb{E}[Q^{(xv)}] = Ad_{xv}^* + \mathcal{O}(N\Delta t^2), \quad (10)$$

where c^* and $d_{(\cdot)}^*$ are the true library coefficients of b_v , a^{vv} , a^{xx} , and a^{xv} , and the $\mathcal{O}(N\Delta t^2)$ remainders are the finite-step drift-squared biases quantified in Theorem 2.3.

Proof. Fix n . Since $K_j(v_{t_n})$ is a deterministic function of v_{t_n} , it is $\mathcal{F}_{t_n}$ -measurable, and $(\xi_n^x, \xi_n^v) \perp \mathcal{F}_{t_n}$ by the Euler–Maruyama construction. For the drift target,

$$\mathbb{E}[K_j(v_{t_n}) \xi \sqrt{v_{t_n}} \xi_n^v | \mathcal{F}_{t_n}] = K_j(v_{t_n}) \xi \sqrt{v_{t_n}} \mathbb{E}[\xi_n^v | \mathcal{F}_{t_n}] = 0, \quad (11)$$

using $\mathbb{E}[\xi_n^v | \mathcal{F}_{t_n}] = \mathbb{E}[\xi_n^v] = 0$. The tower property then gives unconditional zero mean, so $\mathbb{E}[B_j^{(b)}] = \sum_n K_j(v_{t_n}) b_v(v_{t_n}) \Delta t$, which is the drift design equation. For each quadratic target, expand the square (or cross-product) of the increments as in Section 2.5: every stochastic cross-term contains a factor ξ_n^x or ξ_n^v (or both) with the remaining factors $\mathcal{F}_{t_n}$ -measurable, so it vanishes in conditional expectation by the same argument — the correlation ρ between ξ_n^x and ξ_n^v only affects $\mathbb{E}[\xi_n^x \xi_n^v | \mathcal{F}_{t_n}] = \rho \neq 0$, which is a *signal* term (it produces the $\rho \xi v_{t_n} \Delta t$ leading contribution of $Q_j^{(xv)}$), not a source of bias. Summing over n and taking expectations yields the stated identities, with the remaining $\mathcal{O}(N\Delta t^2)$ terms coming from the drift-squared contributions identified in Theorem 2.3. $\square$

Theorem 2.2 (Strong consistency). *Under Assumptions H1–H2, as $T \rightarrow \infty$ with Δt fixed,*

$$\frac{1}{N} A_{jk} \xrightarrow{a.s.} \bar{A}_{jk} := \int K_j(v) \Theta_k(v) \pi(v) dv, \quad \frac{1}{N} B_j^{(b)} \xrightarrow{a.s.} \int K_j(v) b_v(v) \pi(v) dv. \quad (12)$$

If $\bar{A}$ has full column rank, then $\hat{c} \xrightarrow{a.s.} c^$, and analogously $\hat{d}_{vv}, \hat{d}_{xx}, \hat{d}_{xv} \xrightarrow{a.s.} d_{vv}^*, d_{xx}^*, d_{xv}^*$ up to the deterministic $\mathcal{O}(\Delta t)$ discretization floor quantified in Theorem 2.3.*

Proof. By Assumption H1 the chain $\{v_{t_n}\}$ is geometrically ergodic with invariant law π , so Birkhoff’s ergodic theorem applied to the bounded, Lipschitz function $K_j(\cdot) \Theta_k(\cdot)$ gives $(1/N) \sum_n K_j(v_{t_n}) \Theta_k(v_{t_n}) \rightarrow \int K_j \Theta_k d\pi$ a.s., and multiplying by Δt (with $N\Delta t = T$ fixed per unit ergodic average) yields the stated convergence of A_{jk} . Decomposing $B_j^{(b)}$ into a drift term (which converges by the same ergodic argument) and a martingale noise term $\sum_n K_j(v_{t_n}) \xi \sqrt{v_{t_n}} \xi_n^v \sqrt{\Delta t}$ whose predictable quadratic variation grows as $\mathcal{O}(N\Delta t) = \mathcal{O}(T)$, the martingale strong law of large numbers gives that the noise term divided by N vanishes a.s. Continuity of $\hat{c} = (\bar{A}^\top \bar{A})^{-1} \bar{A}^\top \bar{B}$ in $(\bar{A}, \bar{B})$ completes the drift argument; the diffusion arguments for $Q^{(vv)}, Q^{(xx)}, Q^{(xv)}$ follow identically once the $\mathcal{O}(\Delta t^2)$ per-step bias terms of Theorem 2.3 are absorbed into a deterministic remainder that does not grow with T . $\square$

2.5 Finite-Step Bias in the Four Weak Targets

Squaring or cross-multiplying the Euler–Maruyama increments at finite Δt introduces a systematic, non-random contribution from the drift terms, in direct analogy to the scalar diffusion case of [14]. Expanding all three second-moment targets to $\mathcal{O}(\Delta t^2)$ and taking conditional expectations given $\mathcal{F}_{t_n}$:

Theorem 2.3 (Finite-step bias decomposition). *Under Assumption H2, for every n ,*

$$\mathbb{E}[(\Delta v_n)^2 | \mathcal{F}_{t_n}] = a^{vv}(v_{t_n}) \Delta t + b_v(v_{t_n})^2 \Delta t^2, \quad (13)$$

$$\mathbb{E}[(\Delta x_n)^2 | \mathcal{F}_{t_n}] = a^{xx}(v_{t_n}) \Delta t + b_x(v_{t_n})^2 \Delta t^2, \quad (14)$$

$$\mathbb{E}[\Delta x_n \Delta v_n | \mathcal{F}_{t_n}] = a^{xv}(v_{t_n}) \Delta t + b_x(v_{t_n}) b_v(v_{t_n}) \Delta t^2, \quad (15)$$

where $a^{vv}(v) = \xi^2 v$, $a^{xx}(v) = v$, and $a^{xv}(v) = \rho \xi v$. All cross terms linear in ξ_n^x or ξ_n^v vanish by Theorem 2.1; the term quadratic in $\xi_n^x \xi_n^v$ contributes the leading-order signal $\rho \xi v_{t_n} \Delta t$ to (15) through $\mathbb{E}[\xi_n^x \xi_n^v | \mathcal{F}_{t_n}] = \rho$, while the remaining $\mathcal{O}(\Delta t^2)$ terms are the drift-squared biases.

Proof. Substitute the Euler–Maruyama increments and expand each square or cross-product into three terms: a pure-drift term of order Δt^2 , a linear-in-noise term of order $\Delta t^{3/2}$, and a pure-diffusion term of order Δt . The linear-in-noise terms have conditional mean zero by Theorem 2.1. For (15), the pure-diffusion term is $\xi v_{t_n} \Delta t \xi_n^x \xi_n^v$, whose conditional expectation is $\rho \xi v_{t_n} \Delta t$ because (ξ_n^x, ξ_n^v) are jointly standard normal with correlation ρ , independent of $\mathcal{F}_{t_n}$. $\square$

Summing (13) over n against $K_j(v_{t_n})$ gives $\mathbb{E}[Q_j^{(vv)}] = \sum_k d_k^{vv} A_{jk} + \sum_n K_j(v_{t_n}) b_v(v_{t_n})^2 \Delta t^2$, which is the correction implemented in Section 2.6: the already-fitted $\hat{b}_v(v) = \hat{\kappa}(\hat{\theta} - v)$ is used to subtract $\sum_n K_j(v_{t_n}) \hat{b}_v(v_{t_n})^2 \Delta t^2$ before the diffusion regression, following the same two-step order of operations (drift, then bias-corrected diffusion) as [14]. The biases in (14) and (15) are of the same $\mathcal{O}(\Delta t^2)$ order but are numerically negligible at the daily step used throughout this paper: with $\Delta t = 1/252$ and $|b_x(v)|, |b_v(v)| \mathcal{O}(10^{-1})$ – $\mathcal{O}(10^0)$ at plausible variance levels, $b_x(v)^2 \Delta t^2$ and $b_x(v) b_v(v) \Delta t^2$ are of order 10^{-6} – 10^{-5} , several orders of magnitude below the $\mathcal{O}(\Delta t) = \mathcal{O}(10^{-3})$ diffusion signal $a^{xx}(v) \Delta t$ or $a^{xv}(v) \Delta t$. This is why the explicit correction is applied only to $Q^{(vv)}$, where $b_v(v)^2$ can be comparable in magnitude to $a^{vv}(v)$ near the boundary of the state range: the correction is a consequence of Theorem 2.3, not an ad hoc choice.

Proposition 2.4 (Relative statistical efficiency of drift and diffusion recovery). *Under Assumptions H1–H2, for fixed span T the diffusion-type estimators $\widehat{\xi}, \widehat{\rho}, \widehat{\rho\xi}$ attain relative error $\mathcal{O}(\sqrt{\Delta t/T})$ as $\Delta t \rightarrow 0$ (in-fill consistency of quadratic variation), whereas the drift-type estimators $\widehat{\kappa}, \widehat{\theta}$ attain relative error $\mathcal{O}(1/\sqrt{T})$ that does not improve as $\Delta t \rightarrow 0$ at fixed T (long-span consistency of the mean-reversion parameters) [17, 6, 3].*

Proof sketch. The diffusion target $Q_j^{(vv)}$ accumulates $N = T/\Delta t$ approximately independent contributions of order Δt each; by the martingale CLT (Theorem 2.5 below) its fluctuation about the ergodic mean is $\mathcal{O}(\sqrt{N} \Delta t) = \mathcal{O}(\sqrt{T \Delta t})$ against a signal of order $N \Delta t = T$, giving relative error $\mathcal{O}(\sqrt{\Delta t/T}) \rightarrow 0$ as $\Delta t \rightarrow 0$ for any fixed T . The drift target $B_j^{(b)}$ accumulates signal of order T against a martingale noise term whose quadratic variation grows as $\mathcal{O}(N \Delta t) = \mathcal{O}(T)$, giving fluctuation $\mathcal{O}(\sqrt{T})$ and relative error $\mathcal{O}(1/\sqrt{T})$ regardless of Δt : refining the sampling frequency adds more terms to the same finite T -window of mean-reversion information rather than shrinking the estimation-theoretic floor. This asymmetry — classical for diffusion processes observed at high frequency over a finite horizon [29, 4, 5] — is the theoretical counterpart of the empirical pattern in Table 2, where ξ , ρ , and $\rho\xi$ are recovered to under 2% median error while κ and θ require an order of magnitude more data to reach the same precision. $\square$

Theorem 2.5 (Asymptotic normality). *Under Assumptions H1–H2, as $T \rightarrow \infty$,*

$$\sqrt{T} \left(\frac{1}{N} B_j^{(b)} - \bar{B}_j^{(b)} \right) \xrightarrow{d} \mathcal{N}(0, V_j), \quad V_j = \sum_{\ell=-\infty}^{\infty} \text{Cov}[K_j(v_{t_0}) \Delta v_0, K_j(v_{t_\ell}) \Delta v_\ell], \quad (16)$$

with $V_j < \infty$ by geometric ergodicity (Assumption H1); the same statement holds for $Q^{(vv)}, Q^{(xx)}, Q^{(xv)}$ with the corresponding autocovariance sums. Consequently the standard errors of $\widehat{\kappa}, \widehat{\theta}, \widehat{\xi}$, and $\widehat{\rho}$ all decay at the parametric rate $T^{-1/2}$, with the proportionality constants governed by Proposition 2.4.

Proof. Identical in structure to the scalar martingale CLT argument of [14]: decompose each target into an ergodic drift-type sum, which obeys the Markov-chain CLT under geometric ergodicity, and a martingale-difference noise sum, which obeys the martingale CLT with predictable quadratic variation $\mathcal{O}(N \Delta t) = \mathcal{O}(T)$; summing the two limiting variances gives V_j , which is finite because the autocovariances of the geometrically ergodic chain $\{v_{t_n}\}$ decay geometrically and are therefore absolutely summable. $\square$

2.6 Weak Form Kernels

For observations (x_n, v_n) separated by Δt , let $\Delta x_n = x_{n+1} - x_n$ and $\Delta v_n = v_{n+1} - v_n$. For Gaussian kernels

$$K_j(v) = \exp \left[-\frac{(v - c_j)^2}{2h^2} \right], \quad (17)$$

and library $\Theta(v) = [1, v]$, the shared design matrix is

$$A_{jk} = \sum_n K_j(v_n) \Theta_k(v_n) \Delta t. \quad (18)$$

Every kernel defines a soft neighborhood in variance space. Within each neighborhood we use four targets: variance increments for drift, squared variance increments for variance diffusion, squared price increments for price diffusion, and cross-products of the two increments for leverage:

$$B_j^{(b)} = \sum_n K_j(v_n) \Delta v_n, \quad B_j^{(vv)} = \sum_n K_j(v_n) (\Delta v_n)^2, \quad (19)$$

$$B_j^{(xx)} = \sum_n K_j(v_n) (\Delta x_n)^2, \quad B_j^{(xv)} = \sum_n K_j(v_n) \Delta x_n \Delta v_n. \quad (20)$$

The construction is coupled through $(\Delta x_n, \Delta v_n)$ but localized only in the variance coordinate. This choice is deliberate: all Heston coefficients of interest are functions of v , while x enters through its increments. The method is therefore not a general two-dimensional kernel regression over (x, v) . Following Theorem 2.3, the $\mathcal{O}(\Delta t^2)$ fitted-drift contribution $\sum_n K_j(v_{t_n}) \widehat{b}_v(v_{t_n})^2 \Delta t^2$ is subtracted from the $B^{(vv)}$ target before the diffusion regression; the analogous biases in $B^{(xx)}$ and $B^{(xv)}$ are left uncorrected because Section 2.5 shows they are three to four orders of magnitude below the leading diffusion signal at the daily step used here. The fitted linear coefficients give

$$\widehat{\kappa} = -\widehat{b}_1, \quad \widehat{\theta} = \widehat{b}_0 / \widehat{\kappa}, \quad \widehat{\xi} = \sqrt{\max(\widehat{a}_1^{vv}, 0)}. \quad (21)$$

Two leverage normalizations must be distinguished:

$$\hat{\rho}_H = \frac{\hat{a}_1^{xv}}{\hat{\xi}}, \quad \text{imposing the Heston restriction } a_1^{xx} = 1, \quad (22)$$

$$\hat{\rho}_{\text{corr}} = \frac{\hat{a}_1^{xv}}{\sqrt{\hat{a}_1^{xx}\hat{a}_1^{vv}}}, \quad \text{using all three fitted covariance slopes.} \quad (23)$$

These two quantities answer different questions. Only $\hat{\rho}_{\text{corr}}$ is a fitted covariance correlation when the price-diffusion slope is estimated. We also retain $\hat{\rho}_H$ because it reproduces the original empirical -0.33 result under the standard Heston normalization $a_1^{xx} = 1$. The cross-diffusion slope $\hat{\rho}_{\xi} = \hat{a}_1^{xv}$ is reported directly in the noise ablation because it avoids compounding the cross-variation error with error in $\hat{\xi}$.

2.7 Numerical Invariants

Table 1 lists the settings used for every primary estimate. Holding these choices fixed is important because it makes differences across phases attributable to the data or experiment, rather than to a change of estimator. The targeted kernel diagnostic deliberately varies its displayed localization settings, and the nonlinear falsification deliberately adds v^2 to the library. This removes the earlier ambiguity caused by switching among OLS, ridge, LASSO, shifted states, and different kernel settings.

Table 1: Invariant recovery configuration.

Component	Setting
State/increment alignment	contemporaneous, unshifted
Primary library	$[1, v]$
Gaussian kernels	50 centers, 5th–95th state percentiles
Bandwidth	$1.5 \text{sd}(v)/\sqrt{50}$
Regression	LASSO, no fitted intercept
Penalty selection	five-fold cross-validation over 100 penalties
Preconditioning	Euclidean normalization of design columns
Phase 1 reference design	internal step 10^{-4} year; daily sampling; 100 years
Empirical smoothing	one-sided EWMA, span 14

2.8 Indexing and interpretation

The indexing convention fixes what the empirical coefficient means. No state shift is used. In all recoveries, v_n weights the increments from index n to $n + 1$. For the empirical OHLC applications, the day- n proxy uses day- n high, low, open, and close information, while the price series is indexed by daily opens. Consequently the empirical fit is not a causal forecasting regression available at the beginning of day n . It is a descriptive contemporaneous cross-variation convention. This explicit interpretation is essential: shifting the proxy changes the estimand and, in the S&P experiment, removes the retained leverage estimate.

2.9 Generated states and errors in variables

The variance state in market data is generated rather than observed. If an empirical proxy satisfies $\tilde{v}_n = v_n + \eta_n$, then

$$(\Delta \tilde{v}_n)^2 = (\Delta v_n)^2 + 2\Delta v_n \Delta \eta_n + (\Delta \eta_n)^2. \quad (24)$$

Even for independent, mean-zero measurement errors, the last term contributes approximately $2 \text{Var}(\eta)$ to the un-weighted quadratic increment. In addition, using $\tilde{v}_n$ inside the kernels mislocates observations in state space. LASSO and cross-validation do not correct either effect. This is why a smoother can reduce visible proxy noise and still bias the recovered generator. This measurement-error bias is structurally different from — and additive to — the finite-step drift-squared bias of Theorem 2.3: the latter is $\mathcal{O}(\Delta t^2)$, vanishes as $\Delta t \rightarrow 0$, and is removed by the two-step correction of Section 2.5, whereas the $2 \text{Var}(\eta)$ term above does not vanish with Δt and is not addressed by that correction. The comparison below therefore evaluates state construction and attenuation; it is not presented as a formal errors-in-variables correction.

3 Experiments and results

The experiments move from controlled synthetic data to market proxies. Phase 1 checks the estimator when the variance state is observed. Phase 2 corrupts that same state with measured amounts of noise. Phase 3 constructs variance from OHLC prices, and Phases 4–7 test which conclusions survive across market data, repeated simulations, a nonlinear alternative, and different proxies. This ordering separates estimator error from state-construction error.

3.1 Phase 1: multi-path latent-state recovery

The first question is whether the fixed pipeline can recover the Heston coefficients when the correct latent variance is available. We simulate the model by Euler–Maruyama at an internal step of 10^{-4} year and observe the latent log price and variance at daily intervals for 100 years. This is the observation design of the original notebook, with one correction: daily observation times are rounded on the cumulative fine grid instead of truncating every day to 39 fine steps. This removes the inherited clock drift without changing the estimator. The generating parameters are $(\kappa, \theta, \xi, \rho) = (2, 0.04, 0.3, -0.7)$. We repeat this exact fine-step design for 30 independent seeds. Table 2 reports the resulting distribution rather than selecting one favorable path.

Table 2: Latent-state recovery across 30 independent fine-step paths.

Parameter	Truth	Median estimate	10th–90th estimate	Median error%
κ	2.000	1.852	[1.540, 2.262]	12.29%
θ	0.0400	0.04034	[0.03653, 0.04531]	6.61%
ξ	0.300	0.2973	[0.2946, 0.3000]	0.90%
ρ_{corr}	−0.700	−0.7003	[−0.7121, −0.6941]	0.53%
$\rho\xi$	−0.210	−0.2064	[−0.2135, −0.2040]	1.80%

Every seed is below 5% error for ξ , ρ_{corr} , and $\rho\xi$, so the diffusion and leverage result is stable across paths. The same threshold is reached in only 7 of 30 fits for κ and 10 of 30 for θ ; the corresponding worst errors are 32.54% and 21.46%. This gap is exactly the pattern predicted by Proposition 2.4: the diffusion- and leverage-type coefficients ξ , ρ , and $\rho\xi$ are governed by in-fill consistency and improve with sampling frequency at fixed span, whereas κ and θ are governed by long-span consistency and are limited by the 100-year horizon regardless of how finely the path is sampled. Phase 1 therefore supports robust latent-state diffusion and leverage recovery, not uniform sub-5% recovery of all Heston parameters. Figure 1 retains seed 42 only to show the fitted weak functions, while Fig. 2 displays the full 30-seed error distributions.

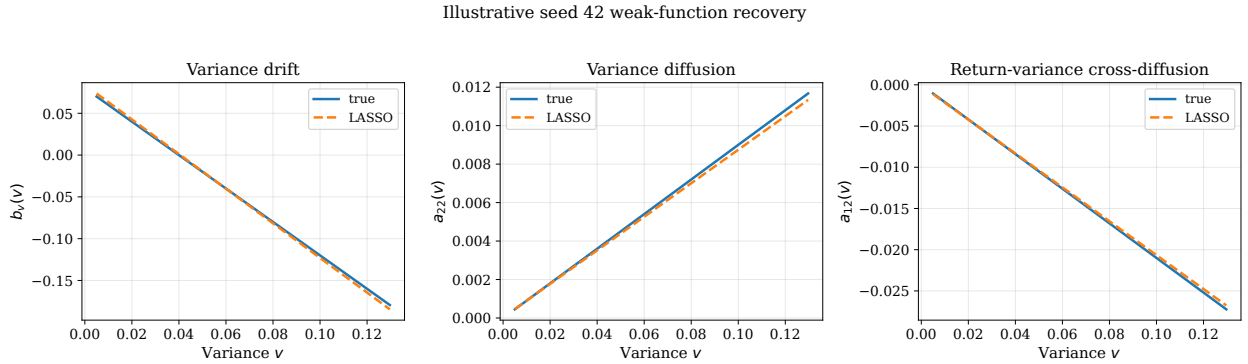


Figure 1: Illustrative seed-42 weak-form fit. This path explains the recovered functions but is not used as the multi-seed robustness summary.

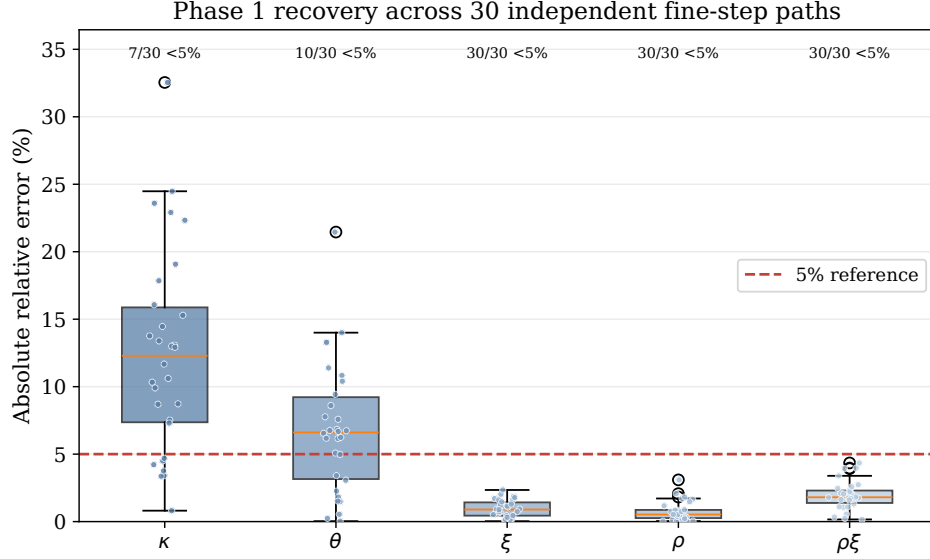


Figure 2: Relative-error distributions across the 30 exact Phase 1 paths. The 5% line separates the uniformly accurate diffusion and leverage quantities from the less stable drift parameters; it is a descriptive reference, not a general empirical acceptance threshold.

3.1.1 Targeted kernel sensitivity

Kernel placement controls which observations are averaged together. The primary configuration remains 50 kernels and bandwidth factor 1.5. As a targeted localization diagnostic, we additionally evaluate 5×5 combinations of 25–100 kernels and bandwidth factors 0.75–3.0 using the illustrative seed-42 path and the same LASSO estimator. Across this grid, the relative error of $\rho\xi$ remains between 1.81% and 2.25%. Drift recovery is more sensitive: κ error ranges from 0.38% to 12.89%, with wider kernels at low kernel counts performing worst. The leverage result is therefore more stable to localization choices than the drift estimate. The full grid in Fig. 3 shows that this path-specific localization statement is not based only on the primary 50-kernel cell. It does not replace the 30-seed variability summarized above.

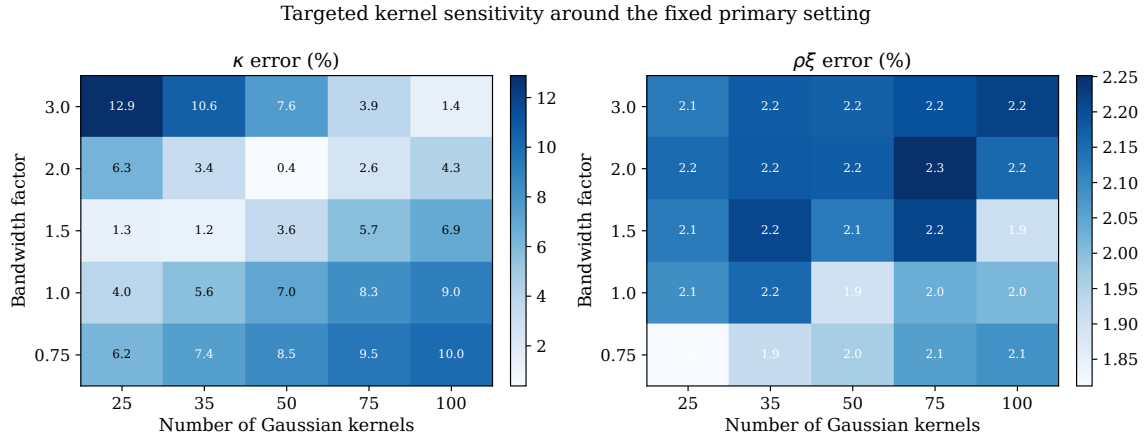


Figure 3: LASSO sensitivity to kernel count and bandwidth. These are targeted diagnostics around the fixed primary setting; they are not alternative pipelines selected after seeing the empirical result.

3.2 Phase 2: leverage robustness to state noise

The next question is how leverage recovery degrades when only the state is corrupted. The price path, true parameters, kernels, and LASSO pipeline are kept fixed, so the experiment isolates state measurement error. The previous

presentation used one curve, emphasized κ , and described a threshold near 32%. We instead evaluate the quantity most directly tied to leverage, $\rho\xi$. For each of 20 noise levels, 100 independently generated Gaussian noise vectors are added to the same reference variance path:

$$\tilde{v}_n = \max\{v_n + \epsilon_n, 10^{-12}\}, \quad \epsilon_n \sim N(0, \gamma^2 \text{sd}(v)^2), \quad (25)$$

where γ ranges from 0 to 0.50. Thus the uncertainty band quantifies measurement-noise draws conditional on one simulated path; it is not a between-path confidence interval. Table 3 gives representative grid points, and Fig. 4 shows all 20 levels and their dispersion.

Table 3: Selected points from the replicated leverage-noise ablation. Errors are relative absolute errors in $\rho\xi$; intervals are the 10th–90th percentiles over 100 noise draws.

Noise SD / state SD	Median error	10th percentile	90th percentile
0.0%	2.12%	2.12%	2.12%
15.8%	4.78%	1.29%	8.16%
31.6%	12.07%	5.70%	18.66%
42.1%	17.37%	8.87%	26.98%
50.0%	22.94%	11.66%	35.50%

The main result is gradual degradation rather than sudden failure. The regenerated results do not establish a sharp phase transition, and the earlier 15% horizontal reference had no external or decision-theoretic justification. At 50% injected noise, a median 7.08% of states are projected back to positivity; the high-noise errors therefore include both measurement corruption and the effect of this declared projection rule.

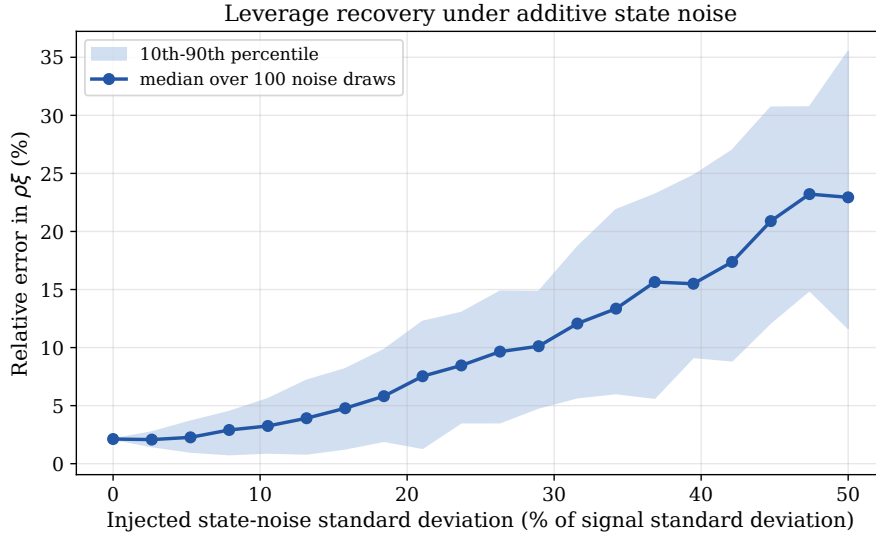


Figure 4: Median leverage-coefficient error with a 10th–90th percentile band. The plotted data and all 2,000 repeat-level estimates are supplied as CSV files.

3.3 Phase 3: one-sided variance-proxy smoothing

The noise ablation starts from an observed state, whereas market applications must build the state from prices. This phase asks whether a one-sided filter can produce a smoother daily variance series without sacrificing the coefficients that we want to recover. Daily OHLC observations are generated from one fixed-seed, 25,000-day Heston simulation with 39 internal steps per day. The filter ranking is therefore a within-path comparison rather than a cross-path guarantee. The raw Garman–Klass variance proxy is

$$\hat{v}_n^{GK} = \frac{1}{\Delta t} \left[\frac{1}{2} (\log H_n - \log L_n)^2 - (2 \log 2 - 1) (\log C_n - \log O_n)^2 \right] [16]. \quad (26)$$

We compare raw GK, EWMA spans 3–60, one-sided rolling means and medians, outlier-clipped EWMA, log-EWMA, and local-level Kalman filters. The predeclared switch rule replaces EWMA-14 only if an alternative has at least 20% lower residual noise-to-signal ratio (NSR) and no larger relative error in either κ or $\rho\xi$.

Raw GK has residual NSR 0.759. EWMA-14 reduces it to 0.371 and raises its correlation with latent variance to 0.952. The lowest NSR is 0.371 for EWMA-14; the closest alternatives, Kalman $q/r = 0.03$ (0.373) and EWMA-10 (0.377), do not achieve the required improvement. EWMA-14 is therefore retained. Figure 5 shows how the retained filter changes the simulated state.

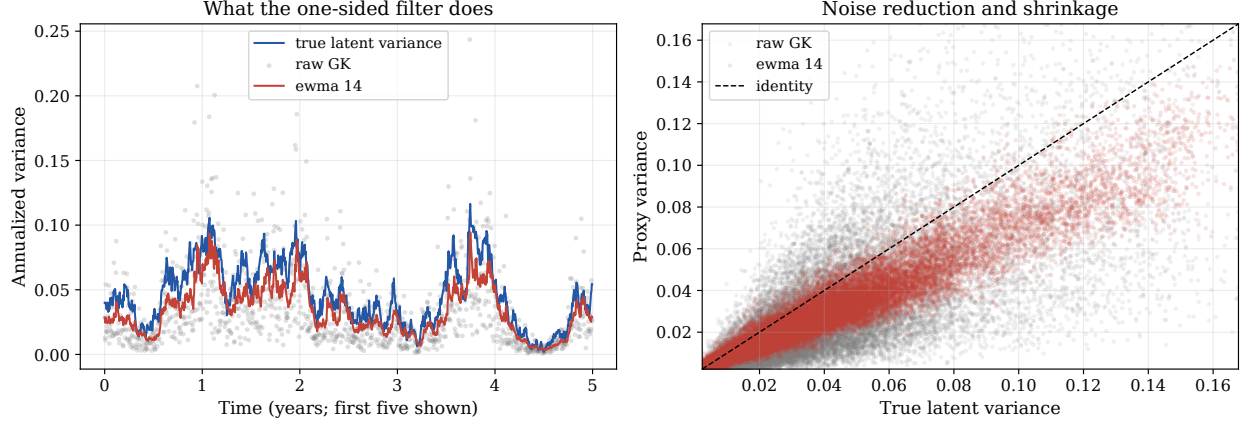


Figure 5: Latent variance, raw GK, and the retained EWMA-14 proxy. The filter is one-sided, and no time shift or future-looking smoother is used.

The trade-off is important. Smoothing lowers high-frequency proxy noise but strongly attenuates the recovered leverage: EWMA-14 gives $\widehat{\rho\xi} = -0.0280$ versus the true -0.210 . Such attenuation is consistent with the known difficulty of estimating leverage from noisy or smoothed volatility proxies [1]. Phase 3 therefore supports EWMA as a continuous, low-noise state construction. It does not support calling the filter an EIV correction or a general solution to proxy bias. Figure 6 places the noise reduction and coefficient-recovery results together so that this limitation is visible.

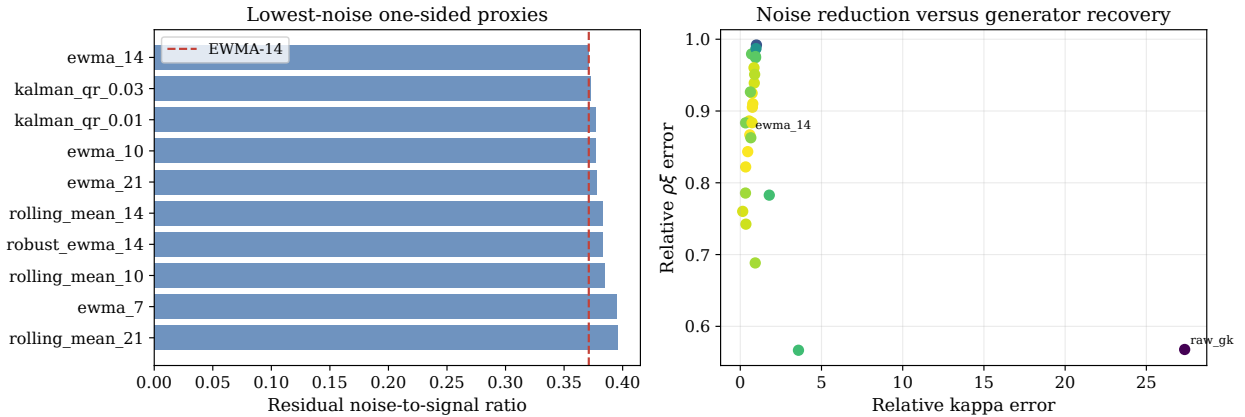


Figure 6: One-sided proxy comparison showing the noise reduction and parameter-recovery trade-off.

3.4 Phase 4: S&P 500, 2007–2010

We now apply the same state construction and estimator to an observed market path. The goal is descriptive: we ask whether the fitted cross-diffusion is negative during a period containing the financial crisis, not whether the fit is a structural or option-pricing calibration. The empirical sample contains 1,007 daily OHLC observations from 3 January 2007 through 30 December 2010. We construct same-day GK variance, apply EWMA-14, and use the invariant unshifted LASSO pipeline. Table 4 reports the slope-derived estimates.

Table 4: Contemporaneous S&P 500 proxy-based estimates.

Sample	$\hat{\kappa}$	$\hat{\theta}$	$\hat{\xi}$	$\hat{a}_1^{xx}$	$\hat{\rho}\xi$	$\hat{\rho}_H$	$\hat{\rho}_{\text{corr}}$
2007–2010	2.361	0.0360	0.530	1.620	−0.174	−0.329	−0.258

The long-run variance estimate corresponds to $\sqrt{\hat{\theta}} = 18.96\%$ annualized physical/historical volatility. It is not an implied-volatility estimate. The mean-reversion half-life $\log(2)/\hat{\kappa}$ is approximately 0.294 years, or 74 trading days. The negative cross-diffusion is consistent with contemporaneous leverage, but the absence of a latent variance ground truth prevents a claim that these are the true Heston parameters. In particular, $\hat{\rho}_H = -0.329$ imposes $a_1^{xx} = 1$, whereas the independently fitted price-diffusion slope is 1.620; using that slope gives $\hat{\rho}_{\text{corr}} = -0.258$. The fitted slopes also fail the Feller condition:

$$\frac{2\hat{\kappa}\hat{\theta}}{\hat{\xi}^2} = 0.604 < 1. \quad (27)$$

Moreover, the affine diffusion fits retain nonzero constants

$$(\hat{a}_0^{vv}, \hat{a}_0^{xx}, \hat{a}_0^{xv}) = (0.00121, -0.00197, 0.00156). \quad (28)$$

The slope-derived quantities should therefore be read as summaries of an affine Heston-type proxy fit, not as an admissible exact Heston calibration. Figure 7 displays the three fitted generator components, while Fig. 8 shows the time-domain proxy and open-to-open returns that enter those weak equations.

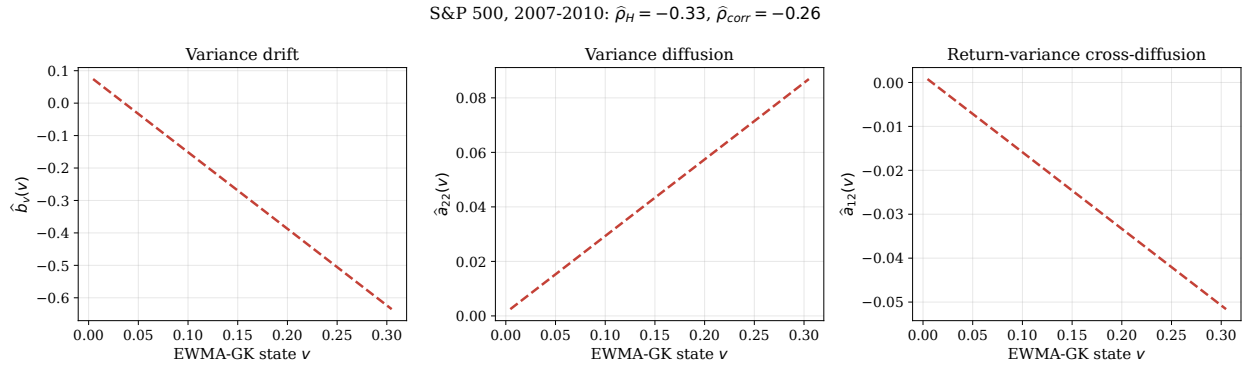


Figure 7: Contemporaneous weak-form functions for the S&P 500 sample.

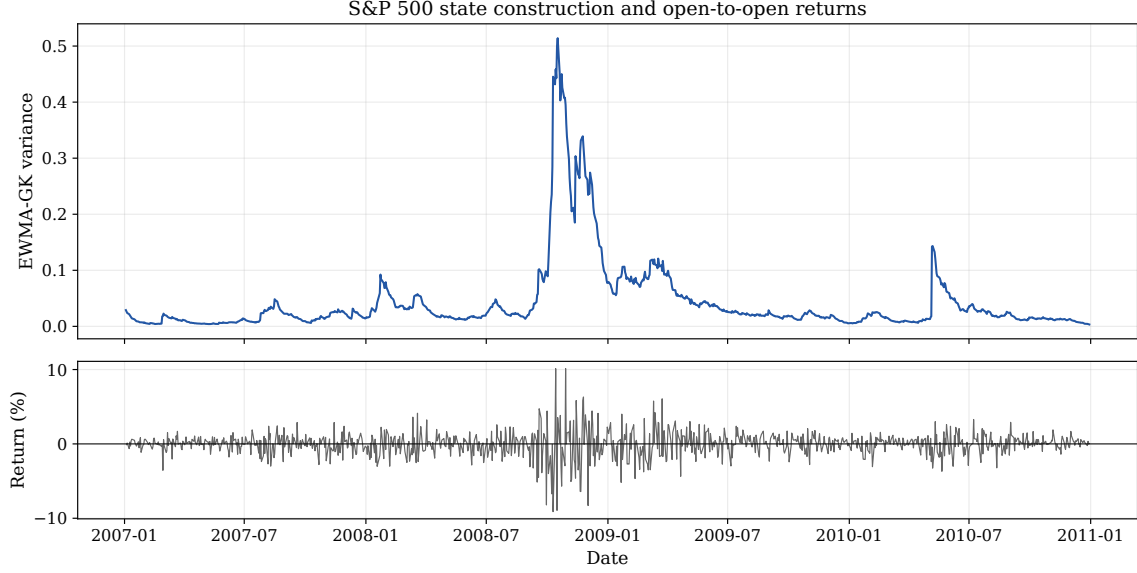


Figure 8: The empirical state construction in time. The variance proxy rises during the crisis while large daily returns cluster. This plot explains the input to the generator fit; it is not a fitted-price or forecasting result.

3.5 Phase 5: multi-seed leverage regimes

One empirical estimate cannot show whether leverage recovery is systematic. We therefore return to latent-state simulation and generate 30 paths at each of four correlations. Every other generating and fitting choice is held constant. Table 5 summarizes the recovered distributions; Fig. 9 shows both their alignment with the identity line and the absolute-error spread.

Table 5: Leverage recovery across 30 seeds. Intervals are the 10th–90th percentiles of $\hat{\rho}$.

True ρ	Median $\hat{\rho}$	10th	90th	Median absolute error
−0.20	−0.200	−0.216	−0.182	0.0079
−0.50	−0.503	−0.515	−0.490	0.0086
−0.65	−0.651	−0.658	−0.639	0.0054
−0.85	−0.850	−0.855	−0.844	0.0033

All 120 latent-state fits recover the correct sign. The result shows that the cross-variation target works across weak and strong negative leverage when the model is matched and the state is observed. It does not remove the empirical proxy problem.

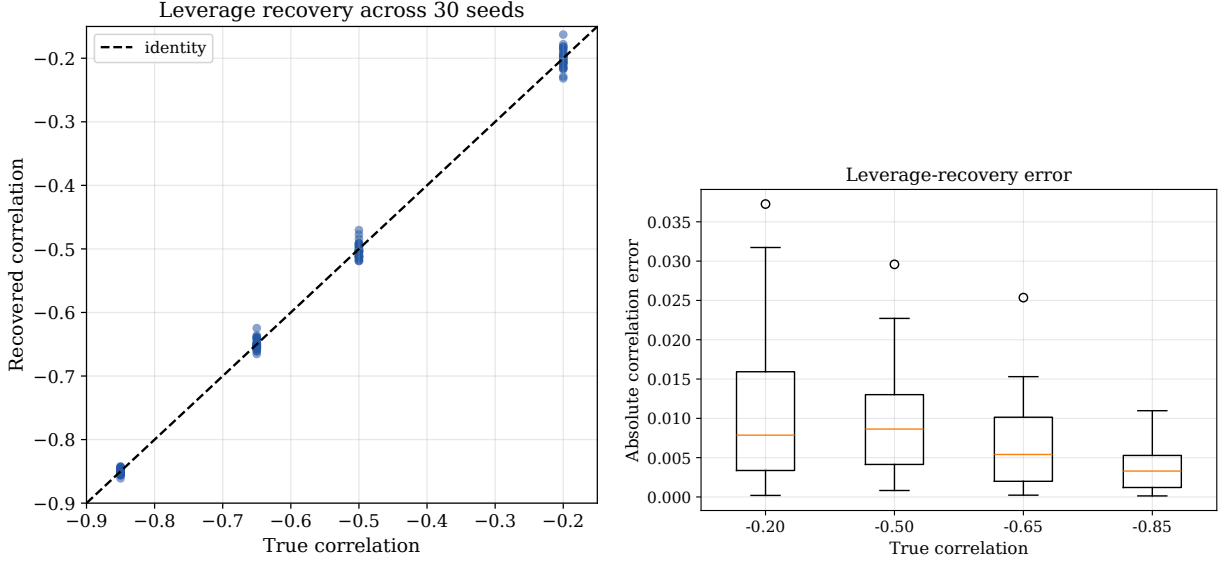


Figure 9: Multi-seed latent-state leverage recovery.

3.6 Phase 6: nonlinear-drift falsification

Good recovery inside a linear Heston library does not show that LASSO can discover a larger nonlinear model. We test that stronger claim directly by adding a quadratic term to the simulated variance drift,

$$b_v(v) = 1.5(0.04 - v) - \beta(v - 0.04)^2, \quad (29)$$

and extend the LASSO library to $[1, v, v^2]$. A useful structure-discovery method should rarely select v^2 when $\beta = 0$. Here it is selected in 28 of 30 null simulations (93.3%). At $\beta = 15$, the median fitted quadratic coefficient is -12.18 , but the 10th–90th percentile interval $[-23.96, 0.93]$ still crosses zero. Selection rates of 93–97% across all tested values are therefore dominated by false positives. “Selected” here means an absolute fitted coefficient greater than the code tolerance 10^{-12} ; no post-LASSO significance test is applied. The useful conclusion is negative: nonzero support is not reliable evidence of a new drift term under this configuration. Figure 10 makes the failure visible in both selection frequency and fitted-coefficient spread.

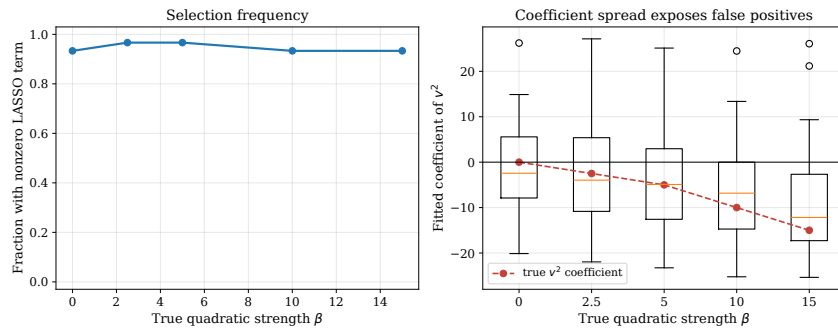


Figure 10: Quadratic selection frequency. The high null selection rate falsifies a reliable nonlinear-support-discovery claim for this configuration.

3.7 Phase 7: Indian-equity proxy sensitivity

The final phase asks a practical question that cannot be answered with one variance proxy: does the sign of recovered leverage remain negative when the same companies are analyzed using different state constructions? The stress test uses a balanced panel of 50 Indian equities from 1 February through 30 June 2020 and five contemporaneous variance proxies: one-minute realized variance, raw GK, EWMA-GK, Parkinson, and Yang–Zhang [2, 25, 30]. Each company–proxy pair

is processed by the same LASSO pipeline. The panel has 100 retained trading days per company (5,000 company-days); every retained company-day contains 316–376 valid one-minute rows. The supplied one-minute Parquet archive is aggregated upstream to the daily proxy panel, and the standardized recovery script starts from that panel.

Table 6: Proxy sensitivity of the recovered leverage sign across 50 companies. Parameter magnitudes are not treated as structural estimates in this short, noisy panel.

Variance proxy	Valid N	Median $\hat{\rho}_{\text{corr}}$	Fraction $\hat{\rho}_{\text{corr}} < 0$
One-minute realized variance	50	−0.320	92%
Raw Garman–Klass	49	−0.424	90%
Parkinson	50	−0.229	80%
EWMA Garman–Klass	41	−0.204	58%
Yang–Zhang	50	0.000	38%

The comparison gives a mixed but interpretable result. Negative leverage is common for the realized-variance, GK, and Parkinson proxies, but it is not universal. Smoothing weakens the sign rate, and Yang–Zhang does not support a panel-wide negative result. The appropriate conclusion is therefore *proxy-sensitive sign evidence*, not proxy-robust identification. Drift and diffusion magnitudes are unstable in this panel—several fits imply nonpositive mean reversion—so they are not given a structural interpretation. “Valid N ” counts finite covariance-normalized correlations; the sign-rate denominator remains all 50 companies, so an undefined correlation is not counted as negative. Table 6 supplies the proxy-by-proxy numbers. Figure 11 retains the failed realized-variance generator as a diagnostic rather than hiding it.

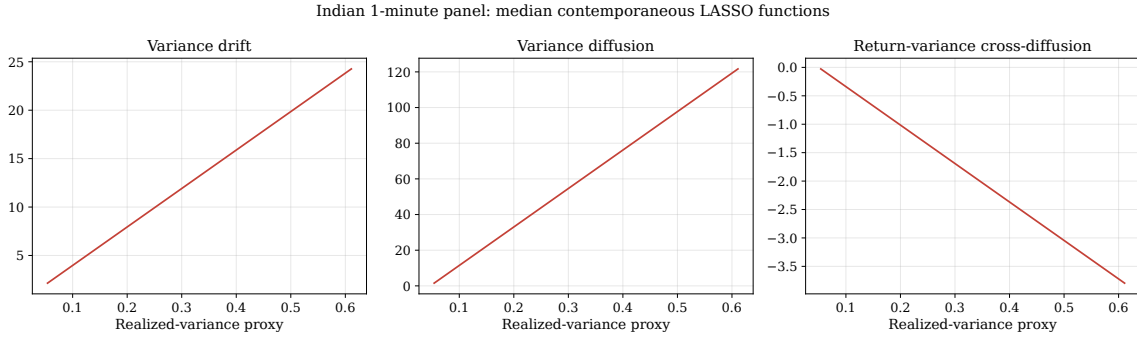


Figure 11: Median fitted drift, variance-diffusion, and cross-diffusion functions for the one-minute realized-variance proxy. The non-mean-reverting median drift is displayed as a failure diagnostic and is not interpreted as a valid Heston generator.

Figures 12 and 13 separate the distribution of finite covariance-normalized correlations from the simpler all-company sign rates. This distinction matters because a zero or undefined LASSO slope is not evidence of a negative relationship.

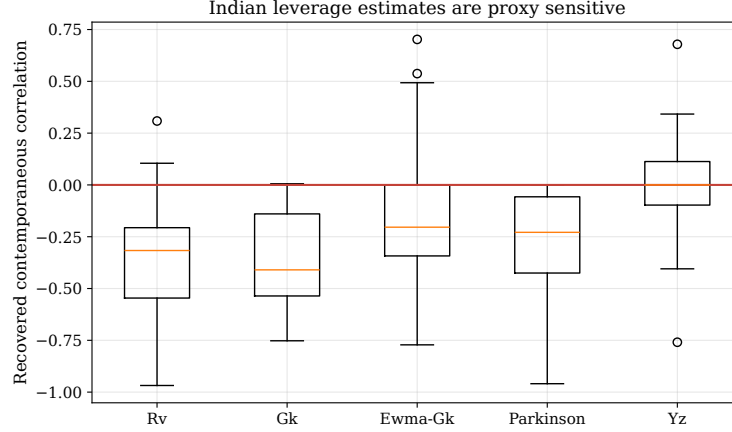


Figure 12: Distribution of admissible contemporaneous correlation estimates by variance proxy.

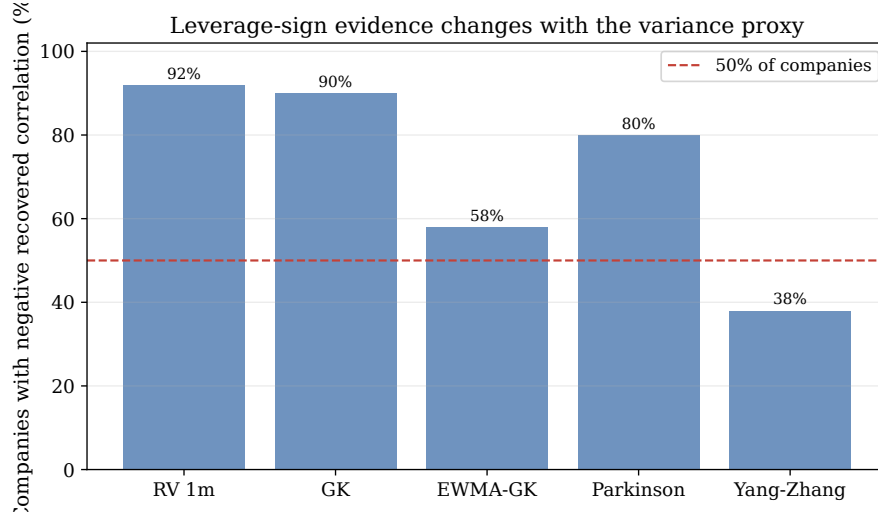


Figure 13: The proxy-dependent conclusion stated directly as a sign rate. Raw realized-variance and range estimators support negative leverage more often than the smoothed or Yang–Zhang proxies.

4 Discussion

The experiments are designed to separate three cases: recovery with an observed state, recovery after controlled state corruption, and recovery from market proxies. The evidence is strong in the first case and progressively more conditional in the other two. This distinction is the main guide for interpreting the results.

4.1 What the experiments support

With an observed latent state and a prespecified Heston library, weak-form LASSO accurately recovers variance diffusion, cross-diffusion, and the leverage correlation. In the 30 exact fine-step Phase 1 paths, all estimates of ξ , ρ , and $\rho\xi$ remain below 5% error. Drift is less stable, with median errors of 12.29% for κ and 6.61% for θ . The separate leverage-regime experiment recovers the correct sign in 120 additional latent-state simulations. The same machinery also reproduces the reported contemporaneous S&P leverage ratio $\hat{\rho}_H = -0.329$ under an explicit Heston normalization; the corresponding fitted covariance correlation is $\hat{\rho}_{\text{corr}} = -0.258$.

The results do not justify stronger claims. Additive state noise gradually degrades leverage recovery; no sharp threshold is established. One-sided smoothing creates a continuous state with lower proxy noise but can attenuate cross-variation substantially. The empirical leverage sign changes with the proxy. Finally, the nonlinear null experiment shows that nonzero LASSO support alone is insufficient evidence of a newly discovered drift term.

4.2 Limitations

The limitations below are not separate from the results; they explain why the synthetic and empirical claims are stated differently.

- **Contemporaneous state construction.** Same-day OHLC uses information accumulated during the day. The empirical estimator is descriptive and cannot be interpreted as an ex ante trading signal.
- **Generated regressor.** The variance proxy is noisy and smoothed. Standard cross-validation does not correct errors in variables, and coefficient uncertainty from state construction is not propagated.
- **Overlapping weak equations.** The five cross-validation folds split 50 kernel equations whose underlying trajectory contributions overlap. The folds are therefore tuning folds, not independent out-of-sample observations; this is especially relevant to the nonlinear false-selection result.
- **One market path.** The S&P sample contains a crisis and only 1,007 daily observations. There is no latent-state truth against which to validate its parameter estimates.
- **Return timing.** The S&P open-to-open return contains an overnight component, whereas same-day Garman–Klass variance is based on the intraday high, low, open, and close. The retained contemporaneous convention does not eliminate this measurement mismatch.
- **Indian sample selection.** The balanced 50-company panel was selected using coverage, liquidity, and sector information from the same stress window. It is a deliberately balanced stress sample; it is neither capitalization weighted nor demonstrably free of survivorship bias.
- **Restricted library.** The main result estimates coefficients inside the Heston form; it does not discover that form without prior specification.
- **Simulation discretization.** Phase 1 uses a fine internal step before daily observation for all 30 paths, whereas the separate leverage regime screen uses direct daily full-truncation Euler for computational replication. The estimator is invariant, but coefficient accuracy should not be compared across these observation designs without accounting for discretization.
- **Uncertainty.** The reported percentile bands describe simulation or injected-noise repetitions, not formal confidence intervals for the empirical generator.

4.3 Next methodological steps

The most important next step is to treat the variance state and the generator as a joint estimation problem. Possible approaches include latent-state filtering, instrumental-variable weak moments, and uncertainty propagation through the proxy construction. The selection problem also needs a stronger validation design, such as separate tuning and evaluation paths or stability selection calibrated under a null model. For higher-frequency data, microstructure noise and asynchronous sampling should be modeled explicitly rather than left to a smoother.

5 Conclusion

The central result is specific rather than uniform. Across 30 synchronized fine-step, daily-observed paths, the standardized weak-form LASSO pipeline keeps ξ , ρ , and $\rho\xi$ below 5% error in every seed. It does not provide the same guarantee for drift: median errors are 12.29% for κ and 6.61% for θ . The separate leverage-regime experiment recovers the correct leverage sign in all 120 paths.

The empirical conclusion is narrower. The contemporaneous S&P 500 fit gives $\hat{\rho}_H = -0.329$ and $\hat{\rho}_{\text{corr}} = -0.258$, but the Indian panel shows that sign frequency depends materially on the variance proxy. The replicated noise ablation explains part of this sensitivity through a gradual error profile, and the smoothing comparison shows that lower proxy noise can come with strong leverage attenuation. The nonlinear falsification further limits the claim to coefficient recovery in a prespecified Heston library. In short, the method identifies leverage well under controlled conditions and provides proxy-sensitive contemporaneous evidence in market data; it does not solve latent-state estimation or general stochastic-structure discovery.

Data and code availability

Code and supplementary materials for this work are available at <https://github.com/Sathvik-Gullipalli/Leverage-recovery-Weak-SINDy>. The repository will be updated with the final reproducibility package upon publication.

Indian panel composition

The balanced panel contains five selected names from each of ten broad sectors. This table restores the sample definition from the previous draft; it does not imply that sector balance makes the 50 firms representative of the full Indian equity market.

Sector	Symbols
Energy and utilities	RELIANCE, POWERGRID, ONGC, NTPC, COALINDIA
Non-bank financials and insurance	BAJFINANCE, BAJAJFINSV, HDFCLIFE, CHOLAFIN, SBILIFE
Banks	HDFCBANK, ICICIBANK, AXISBANK, SBIN, KOTAKBANK
Consumer discretionary, services, telecom	BHARTIARTL, ASIANPAINT, TITAN, INDIGO, ADANIPTS
Automobiles	MARUTI, TMPV, M&M, BAJAJ-AUTO, EICHERMOT
Consumer staples	HINDUNILVR, ITC, NESTLEIND, BRITANNIA, DABUR
Information technology	TCS, INFY, HCLTECH, TECHM, WIPRO
Industrials, capital goods, construction	LT, BEL, SIEMENS, ABB, HAL
Healthcare	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, APOLLOHOSP
Materials, metals, cement	TATASTEEL, ULTRACEMCO, JSWSTEEL, HINDALCO, GRASIM

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